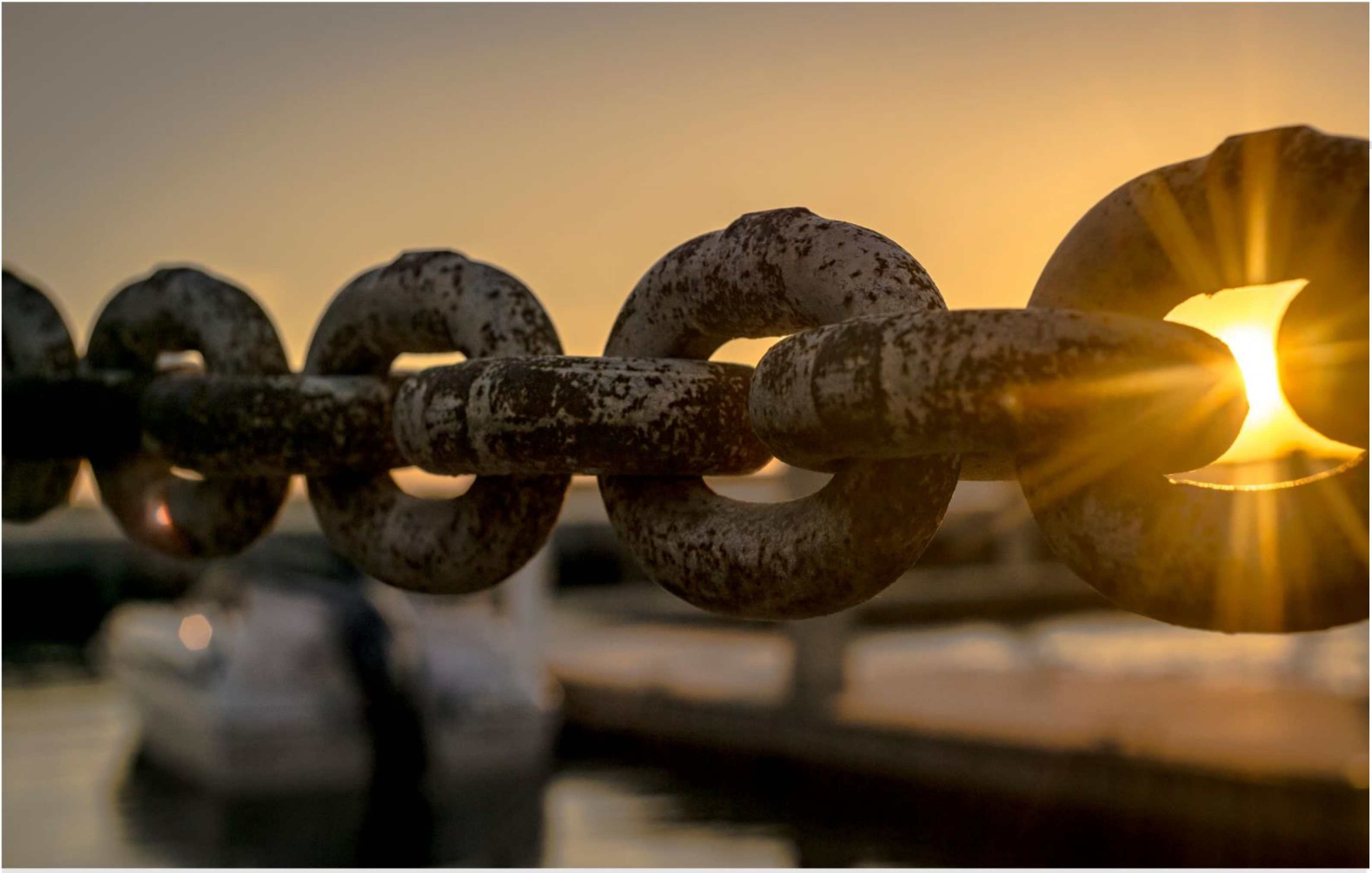


Adroit SCADA

D_COMLI_SER
Communications Link Serial Protocol Driver

Advanced Worx 112 (Pty) Ltd
20 Waterford Office Park
189 Witkoppen Road (c/o Waterford Drive)
Fourways
South Africa
adroittech.co.za

Version	Date	Author	Comment	Approval
14	17/03/2021	David Jordaan	Updated documentation.	
18	16/11/2022	David Jordaan	Shadow mode partner (with offline redundant plc) in standby did not mark affected tags bad when becoming master. For Shadow and Parallel, when switching cluster: unchanging tags were not updated WARNING: Requires at least Rev92 of t_device.dll	

Contents

1. Communications link Serial protocol Driver

1.1. Overview

This driver supports the ABB Communications link driver.

The 2nd Generation driver allows for the following features:

- Improved performance due to optimizations on mixed I/O to the same protocol area being transacted only once removing redundant transactions.
- Serial drivers by design don't benefit from multiple channels thus only one channel per device is enforced. Up to 247 Multi-dropped devices are supported per port.
- Demand scanning support, where read only I/O that is not on a mimic / graphic form and not logged or alarmed will be removed from the scanning system, the applicable tasks will dynamically reduce is size and merge into fewer scan tasks.

For improved network performance and reliability, it is recommended to use of industrial USB serial ports or true serial cards into the machines where applicable. RS232 and RS485 and RS422 is supported Windows traditionally expects com ports to always be available, loosing contact with Ethernet serial extenders can cause lockups… please contact Adroit support if you suspect this.

1.2. Test Hardware

This driver has been tested on the following hardware:

FX5U -32MT/DSS Plc using USER_FB_Comli_FX5U function block dated 2019/04/24 14:25:42

1.3. Wiring

Control of DSR/DTR CTS/RTS signalling is available as needed.
RS-232-c / RS-422 / 485, 4 and 2 wire supported.
Reflection detection is configurable, (future …)

<u>DB9 , DTE to DTE cable diagram RS-232C - 3 wire (With loopbacks in case the system expects signalling)</u>

PLC 9 pin		PC 9 pin
2 RX		2 RX
3 TX		3 TX
5 GND		5 GND

Note: *Connect shield to one side , PLC side only*

DB9 , DTE to DTE cable diagram RS-232C - 3 wire (With CTS/RTS loopbacks in case the system expects signalling, And DSR,DTR pulled high/Low on CD)

PLC 9 pin		PC 9 pin
2 RX		2 RX
3 TX		3 TX
5 GND		5 GND
7 RTS 8 CTS		7 RTS 8 XCTS
1 CD 4 DTR 6 DSR		1 CD 4 DTR 6 DSR

Note: *Connect shield to one side , PLC side only*

Also used for ABB AC-400 series DCS (diagram as above)

DB9 , DTE to DTE cable diagram RS-232C - 7 wire

PLC 9 pin		PC 9 pin
2 RX		2 RX
3 TX		3 TX
5 GND		5 GND
7 RTS 8 CTS		7 RTS 8 CTS
4 DTR 6 DSR		4 DTR 6 DSR

Note: *Connect shield to one side , PLC side only*

RS422/485 – 2 wire , bus cabling

PLC		PC
SG		Signal ground
SDA RDA		R+(D+)
SDB RDB		R-(D-)

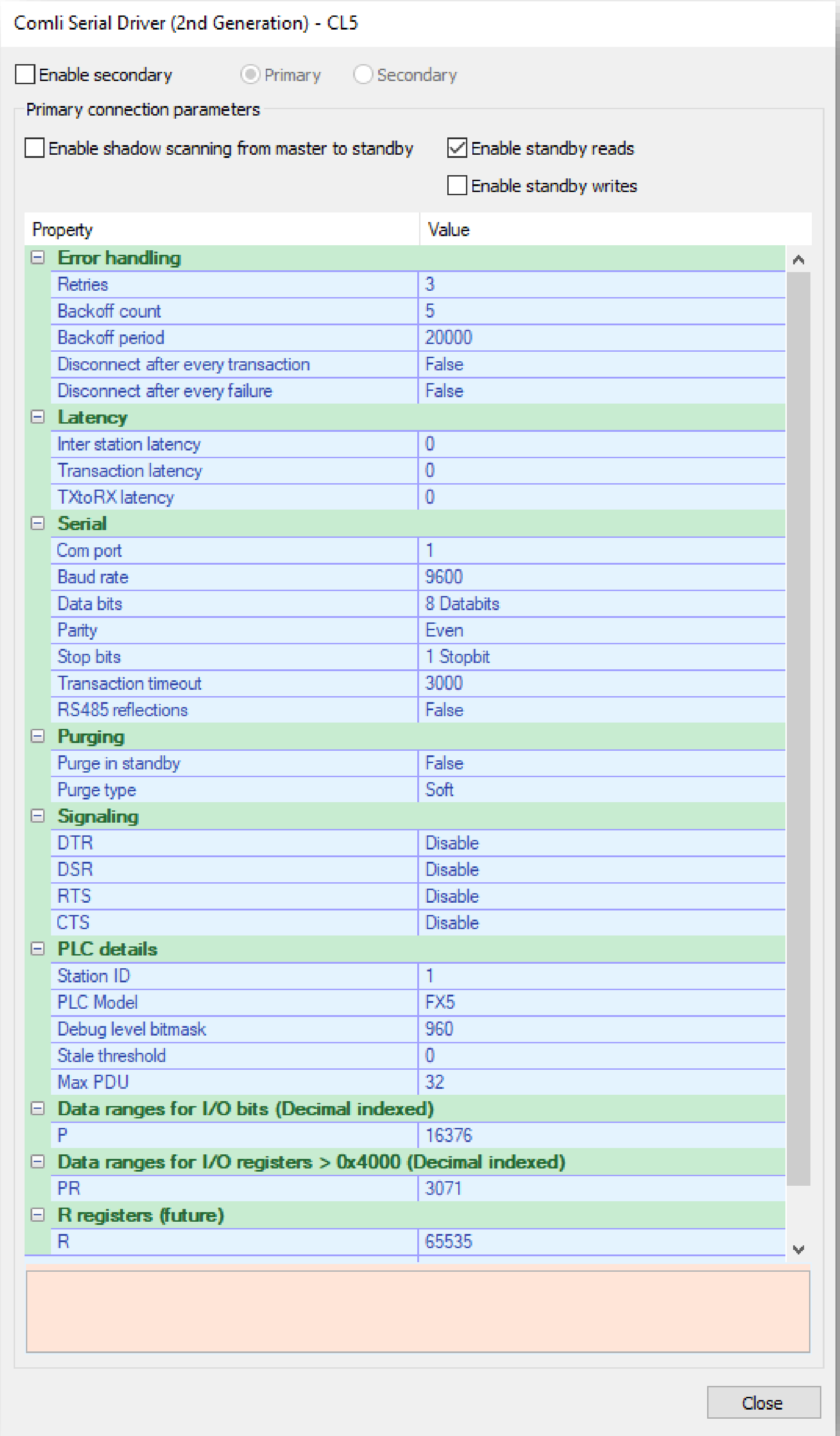
*Polarity is important

RS422/485 – 4 wire , bus cabling

PLC		PC
SG*		Signal ground
SDA		R+
SDB		R-
RDA		T+
RDB		T-

**Ground can be left disconnected*
**Polarity is important*

1.4. Device Configuration



Enable secondary – Turn this option to have the driver maintain 2 channels of communication, the radio buttons **Primary** and **Secondary** can then be used to toggle between configurations.

PLC Details

Station ID – (1 default) PLC station id 1 – 247 , Master is always 0.

PLC Model – (ABB CPU default) – Choose a specific model (future)

Debug level bitmask – (480 default) - this is a bit mask that enable extra debugging in the data-scope output, do not leave it enabled during normal operation on high speed drivers as it will slow down the system, depending on speed requirements, in serial drivers this delay is negligible.
960 default is just the protocol messages 40hex + 80hex+ 100hex + 200hex = 3C0hex -> 960 decimal in the dialogue

```
#define DEBUG_MASK_COMSRESOURCE_INFO         0x00000001
#define DEBUG_MASK_COMSRESOURCE_CRITICAL     0x00000002
#define DEBUG_MASK_PROTOCOL_ITEM_VALUES      0x00000004
#define DEBUG_MASK_PROTOCOL_ITEM_BAD_VALUES  0x00000008
#define DEBUG_MASK_NASA_POOLS                0x00000010
#define DEBUG_MASK_TIMING                    0x00000020
#define DEBUG_MASK_PROTOCOL_RESPONSES        0x00000040
#define DEBUG_MASK_PROTOCOL_REQUESTS         0x00000080
#define DEBUG_MASK_PROTOCOL_READ_DESC        0x00000100
#define DEBUG_MASK_PROTOCOL_WRITE_DESC       0x00000200
```

Stale threshold – **(5 default**) – Every nth occurrence of a unchanged item value will be sent to the master, if the value is different the value will be send regardless of this count. This count reduces unnecessary event withing the agent server. Set to to report every poll value regardless of change , 0 is not a valid option.

Max PDU length – **(32 default)** - The overall maximum transaction length , 1 is the lowest value up to 32 16bit-registers or 32 x 16 discreet bits digitals per transaction.

Serial

Com Port – Only insert the number for the port, E.g. "4" for com4

Baud rate – Combo (9600 default) – select the required baud rate from the selection provided.

Databits – Combo (8 default) – select the required databit count from the selection provided.

Parity – Combo (ODD default) – select the required Parity mode from the selection provided.

Stopbits – Combo (1 default)) – select the required stopbit count from the selection provided.

Transaction timeout – (3000 mSec Default) – insert the required timeout for the system in milliseconds.

485 Reflections (false by default) – check if reflection detection and compensation is required on the port (future …)

Purging

The driver will purge TX and RX buffers (data on the port) by default when required.
However, some systems such as Ethernet to serial port servers can become overloaded or even lockup if left receiving bytes in its internal buffer without a driver reading anything for a long period, for example when a cluster partner is in standby use the setting below to compensate.

Purge port in standby (false by default) – check if purging of the buffers are to be performed in the background while the cluster partner is in standby.

Purge mode – (SOFT default) Soft cause a simple flushing via reading any available bytes, HARD sends a system call to tell the device itself to flush, some systems have become unresponsive in hard mode.

Error handling – Multi-dropped system variables

Retries – Enable this button to have the driver keep the PLC connection alive. Please see the PLC program and heartbeat sections below. This option is only useful with the watchdog program running in the PLC.

Backoff count – (5 default) – require 5 full consecutive failures including retries.
e.g. if 3 retries is configured and the **backoff count** is 5, the system will go into a backoff delay period in which this device will not be polled or written to for the Backoff period, in this example after the ((1+3)*5) consecutive failures.

Backoff period – Specify the millisecond interval the a device will stay silent on the channel, this is very useful in multi-dropped systems as failing devices do not severely affect online devices.

Disconnect after every transaction – (false default) -tell the channel to close the port after every transaction, can be useful on gateways if required.

Disconnect after every error – –(false default) – tell the channel to close and reopen the port on every error during retries else it will only occur after the final retry

Latency

Inter-PLC latency – The period that the system will wait before transmitting any read request or write request in multi-dropped system when a new device is contacted.

Transmission latency – The period that the system will wait before transmitting any read request or write request for the same plc device. Inter message delay.
Tx to RX latency – The period that the system will wait after transmitting a request and before attempting to read the response.

Data ranges

P – I/O bits area , bits below 4000hex index is in bits , E.g. P 6, P 7, … P n (default 16376)

PR – I/O bits above 4000hex (addressed in octal , index is in 16 bit registers , E.g. PR 0 W, PR 1 W … PR n W

As of Rev 13 PR area now supports Bits in Registers as well. E.g., PR 1000.0 the bit index is in decimal. 0 -> 15

R – Registers { assumed these are referring to the high registers }

A – Analogue outputs (future)

AI – Analogue inputs (future)

T 0 – Time Sync can be scanned to any agent slot, but MUST be scan inhibited and output enabled, this is used to send down master time to the outstation.

1.5. Supported Addresses

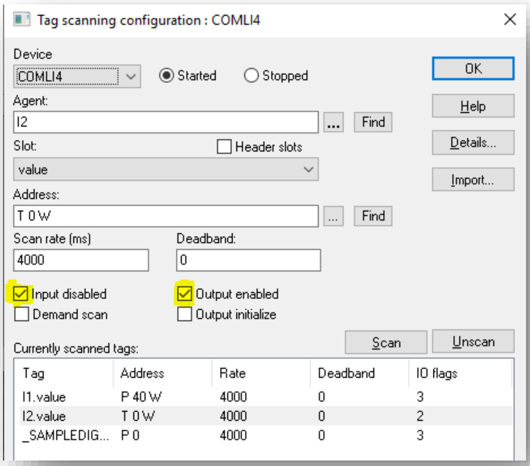
ABB Comli PLC data (device) types supported in the *Adroit* Protocol , note Decimal is use throughout.

Device	prefix	Max Range	PLC Data Type	BOOLEAN	INTEGER	REAL	STRING
P – I/O bits	P	0 - 16376	**Bit**	P 0	-	-	-
(Octal)			**Word16** **Int16**	- -	P 0 W P 0 I	P 10 W P 70 I	- -
PR – I/O registers	PR	0 – 3071	**Bit** *	PR 0.0 PR 0.15 PR 0 W	-	-	-
(Decimal)			**Word16** **Int16**	- -	PR 0 W PR 79 I	PR 1 W PR 89 I	- -
T – Time sync	T	0	**Bit**	T 0	-	-	-
(write only, input inhibited)			**Word16**	-	T 0 W	T 0 W	-
R (Future)	R	0 - 65535	**Int16**	-	-	-	-
			Word16	-	-	-	-
			DWord32	-	-	-	-
			Long32	-	-	-	-
			Float32	-	-	-	-
			Int16	-	-	-	-
			String	-	-	-	-
A AI (Future)	A	-	**Int16**	-	-	-	-
			Word16	-	-	-	-
			DWord32	-	-	-	-
			Long32	-	-	-	-
			Float32	-	-	-	-
			String	-	-	-	-

* PR 0 W as a whole word can be scanned into a digital so that any value is deemed 1(true) and 0 is 0(false).

1.6. Timesync considerations

Scan address T 0, T 0 W scan to any discreet agent type such as BOOLEAN, INTEGER or REAL agent. Make sure the "INPUT DISABLED" and OUTPUT ENABLED" check boxes are checked, as this type is write only.



The driver will then send down the system time in UTC time format.

1.7. Comli – Protocol function supported

0 30H (Write) Transfer Multiple I/O bits (P 0) or register (P 0 W)

1 31H Acknowledge.

2 32H Requests multiple I/O bits or registers groups of at least 8 bits, registers are groups of 16bits

3 33H (Write) Transfer single I/O bit

4 34H Request a single I/O bit. (redundant function 2 is used)

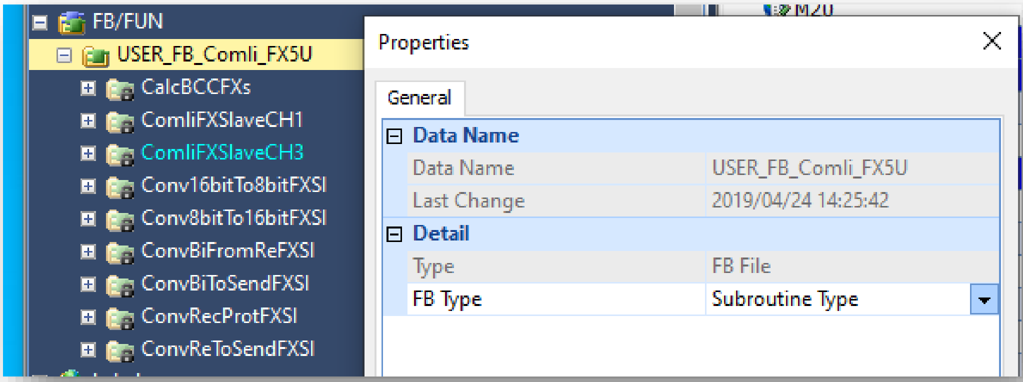
J 4AH Transfer date and time in UTC format.

If more messages are required please contact the Adroit driver department, these will be added as required.

2. Configuring Mitsubishi FX5 as a Comli slave

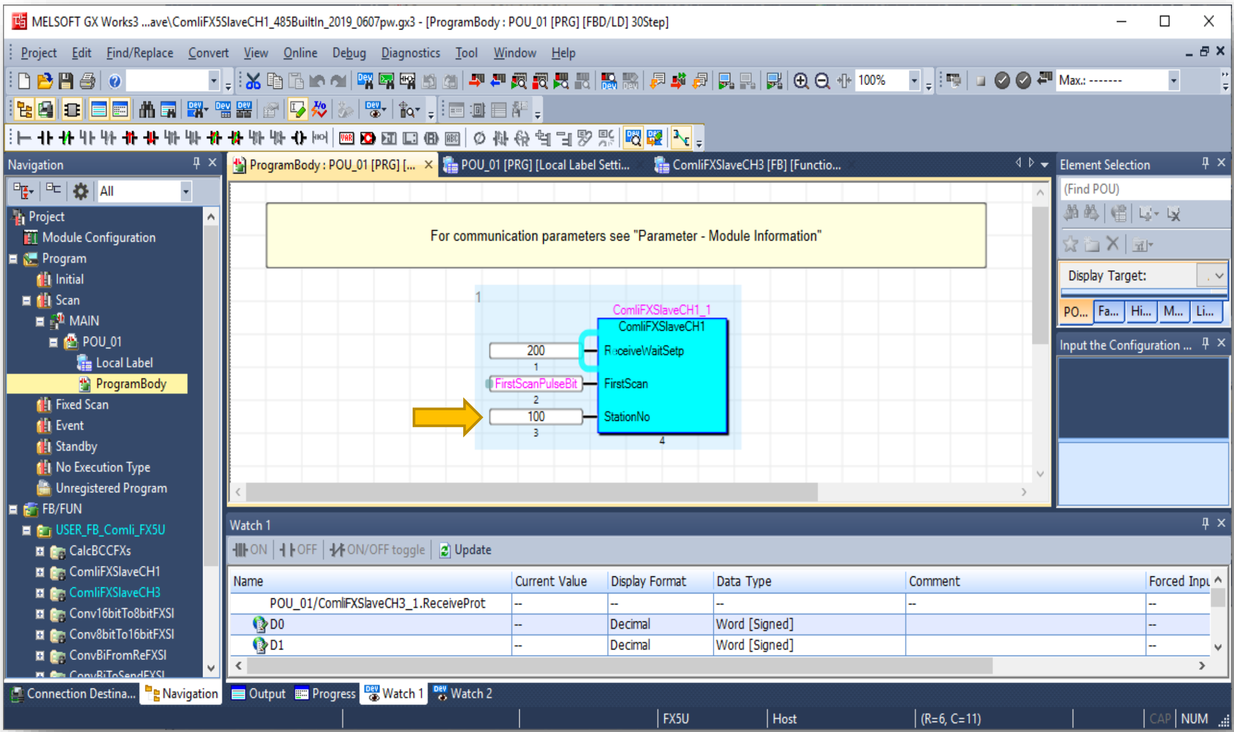
2.1. Comli function block setup

This example uses the FX5U -32MT/DSS Plc using CH1 of USER_FB_Comli_FX5U function block dated 2019/04/24 14:25:42

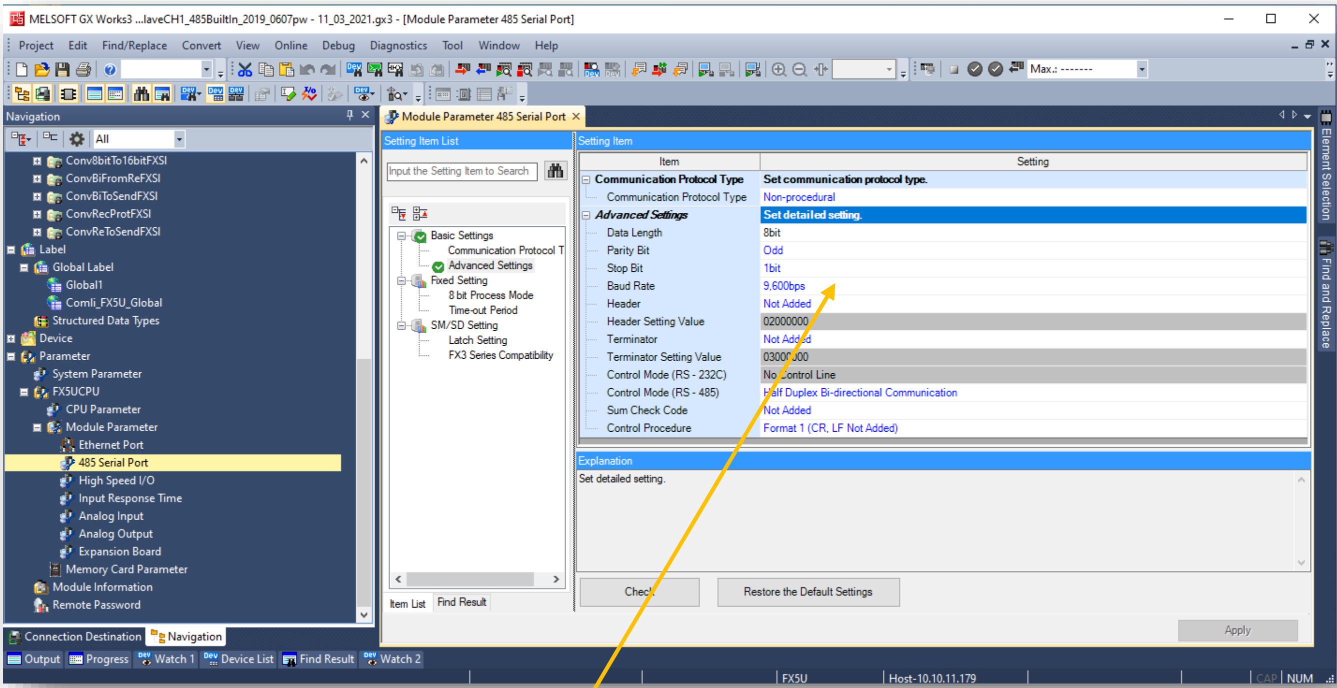


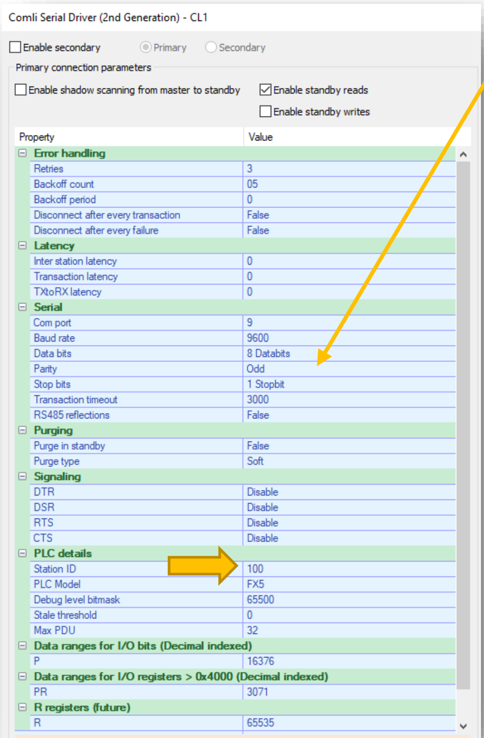
2.2. PLC program

Fx5 plc with a serial port, in this example a FX5U-32MT/DSS was used.
The 485 port on the FX5 is channel 1, therefore ComliFXSlaveCH1 function block must be used.



Make sure the com port settings match the Master driver settings. Baud rate ,Databit length, stopbits and parity. And that the station number matches.





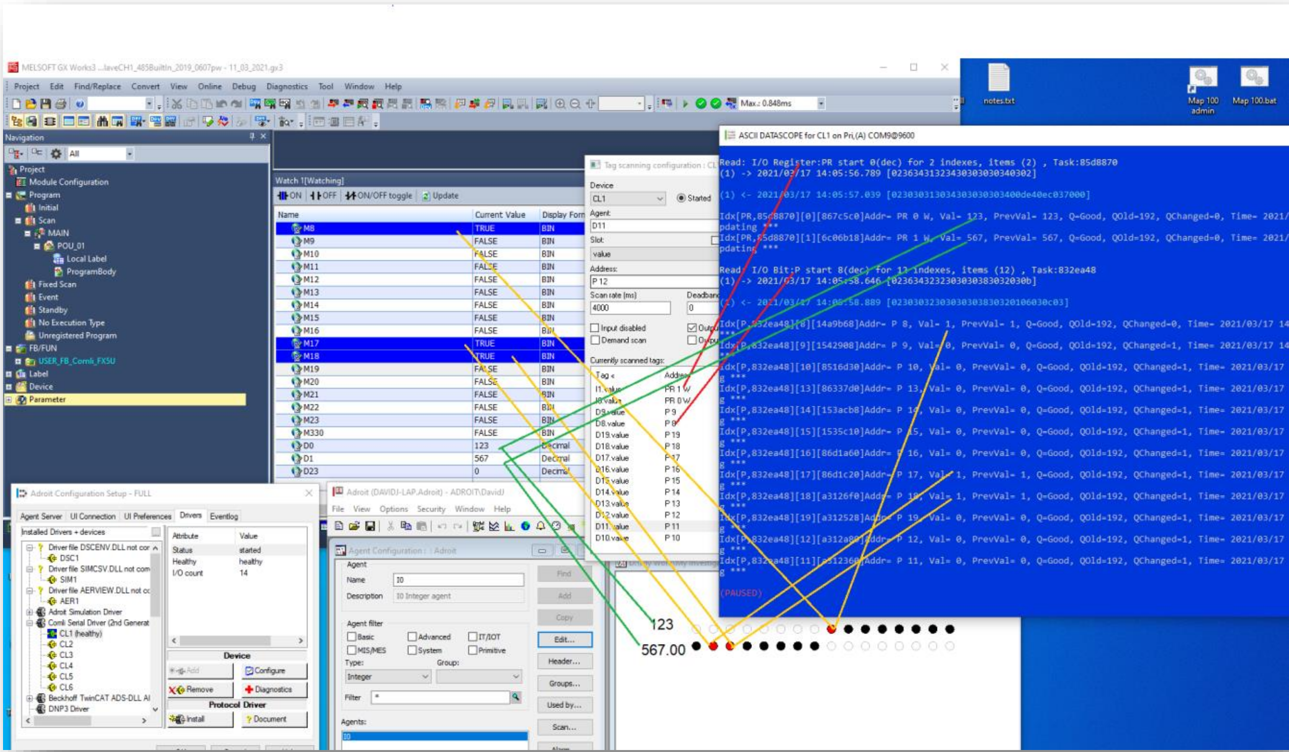
2.3. PLC Wiring

<u>Example provided is wired for RS485 (2 wire)</u>

Hint: use 2 terminal programs and a second RS485 port, make sure to test the link by sending and receiving on the opposite end in both directions. This confirms the TX and RX buffers are working on the 485 modules.

2.4. Resulting output

Resulting montage PLC monitored via gxworks3, Adroit classic UI, Adroit Datascope below.



2.5. FX5 / Comli I/O mapping

- COMLI P's are described in OCTAL in the Comli protocol specification, but we have decided to prevent confusion to rather standardize as decimal.
- P's map to M's in the FX5 plc … P0 ~ M0
- P 0 W in Adroit is a 16bit array or P's mapped to Comli P0, P1 ... P15 ~ [M0, M1 ... M15]
- PR's in comli are any 16bit reg addressing decimal higher than 0x4000 but addressed as decimal starting from 0 such that PR 0 W maps to D0 in Mitsubishi.
- PR 0 W ~ D0 , PR 1 W ~ D1 … etc
- PR 0.0 bit in Register read and writes (with read before write) to digital agents is supported.
- PR 0 W as a whole word can be scanned into a digital also so that any value is 1, 0 is 0.